

TEST BANK

CHAPTER 10

SYMBIOTIC RELATIONSHIPS

Matching

In this section, you are to select from the choices listed in column B the term or phrase that best matches the term or phrase listed in column A and then to place the letter for the selected choice from column B in the space provided to the left of the corresponding item in column A.

[Note to instructors: Here, the choices are correctly matched and thus should be randomly assorted before the exam is prepared.]

Column A		Column B
___ 1. parasitism	_____→	A. one animal lives at the expense of another
___ 2. mutualism	_____→	B. two animals associating together for mutual benefit
___ 3. zooxanthellae	_____→	C. dinoflagellate symbionts
___ 4. cyanellae	_____→	D. cyanobacteria symbionts
___ 5. cryptic symbiosis	_____→	E. host and guest symbionts are completely integrated
___ 6. inquilinism	_____→	F. animals sharing living quarters
___ 7. commensalism	_____→	G. one animal benefits while the other is not harmed in an association
___ 8. epizoites	_____→	H. commensal animals that live on other invertebrates
___ 9. endozoites	_____→	I. commensal animals that live in other invertebrates
___ 10. phoresis	_____→	J. symbionts merely traveling together

Multiple Choice

Circle the choice that best completes or answers the given questions.

[N.B.: The correct answer to each of the questions is indicated by an asterisk. For most of the questions, the letter “a” is the correct choice; reorder the choices as you see fit.]

1. Dinoflagellates and diatoms that have symbiotic relationships with invertebrates are called _____.

a. zooxanthellae *
b. zoochlorellae
c. cyanellae
2. Many tube-dwelling commensal invertebrates exhibit a strong _____.

a. thigmotaxis *
b. inquilinistic relationship with their host
c. chloroplast symbiosis
d. modification in their growth pattern
e. None of the above

176 Test Bank for Marine Biology

3. Algal symbiosis, whether cellular or chloroplast, occurs in all but which one of the following invertebrate phyla?
 - a. Arthropoda *
 - b. Mollusca
 - c. Echiura
 - d. Porifera
 - e. Cnidaria
4. Commensal invertebrates that live on other invertebrates are called _____.
 - a. epizoites *
 - b. endozoites
 - c. cryptic symbionts
 - d. cyanellae
 - e. None of the above
5. Commensal invertebrates that live inside other animals are called _____.
 - a. endozoites *
 - b. epizoites
 - c. cryptic symbionts
 - d. cyanellae
 - e. None of the above
6. Symbiotic relationships between luminescent bacteria and marine animals are generally confined to which of the following groups?
 - a. kelp *
 - b. zooxanthellae
 - c. zoochlorellae
 - d. cyanellae
7. Certain fish are able to live among anemones because of which one of the following features?
 - a. The fish prevent discharge of the nematocysts. *
 - b. The fish are immune to attack from the nematocysts.
 - c. The fish have a thick integument (= skin).
 - d. The fish can avoid the tentacles.
 - e. The fish select special anemones that have no nematocysts.
8. ____ refers to symbiotic association in which one member gains an advantage without harming the other.
 - a. Commensalism *
 - b. Parasitism
 - c. Mutualism
 - d. Inguilism
9. A symbiotic association that is required by both members and in which both members gain an advantage is called _____.
 - a. commensalism
 - b. parasitism
 - c. mutualism *
 - d. inguulism
10. ____ is a type of symbiotic relationship in which an animal lives in the home of another or in its digestive tract.
 - a. Inguilism *
 - b. Mutualism
 - c. Parasitism
 - d. Commensalism
11. The jellyfish *Cassiopeia* satisfies most of its energy needs through _____.
 - a. symbiotic relationships with zooxanthellae *
 - b. symbiotic relationships with zoochlorellae
 - c. predation
 - d. herbivory
 - e. detritus feeding
12. Photosynthetically active cyanobacteria living within sponges translocate about ____ % of the organic carbon they fix to their sponge host.
 - a. 10 *
 - b. 25
 - c. 50
 - d. 75
 - e. 95
13. A symbiotic association in which one member gains an advantage at the expense of the other is called a(n) ____ relationship.
 - a. epizootic *
 - b. endozootic
 - c. parasitic
 - d. None of the above

True/False

Mark each of the given statements with the letter *T* if it is true and *F* if it is false.

[Instructors: In some cases, a few options are included for the same question.]

1. [False] Nearly all tropical sea anemones, soft and stony corals, and sea whips and fans possess zoochlorellae in their tissues.
2. [True] The Portuguese man-of-war provides a kind of sanctuary to small fish that seek shelter among its tentacles.
3. [False] A curious feature about sea anemones, soft and stony corals, and sea whips and fans is that they do not possess symbiotic algae in their tissues.
4. [True] The light produced by bioluminescent bacteria symbiotic within certain deepwater squids and fishes is produced continuously by the bacteria.
5. [False] The giant clam, *Tridacna*, has the same body orientation and external anatomy as any other clam.
6. [True] *Chaetopterus* produces a U-shaped, parchmentlike tube that it shares with pinnotherid crabs.
7. [False] Of the numerous kinds of symbiotic relationships known to occur between marine organisms, none is known to occur between echinoderms and vertebrates.
8. [True] Commensalism refers to a symbiotic association in which one member gains an advantage without harming the other.
9. [True] Sea anemones and crabs have symbiotic relationships, including one in which the anemones provide a type of camouflage for the crabs.
10. [False] Cleaning fishes and shrimps have few distinguishing morphological features.
11. [True] In cleaning symbiosis, the cleaners gain food, while the fishes that are cleaned enjoy improved health.
12. [False] In cleaning symbiosis, the cleaners gain a refuge from predators, while the fishes that are cleaned enjoy the extra food brought to them by the cleaners.
13. [True] The acoel flatworm, *Convoluta*, has zoochlorellae in its epidermal cells.
14. [False] Zooxanthellae that live as symbiotes and zooxanthellae that are cultured as independent free-living organisms exhibit the same morphology.

Short Answer (Fill in the Blank)

1. When the distinction between host cell and algal symbiont becomes indistinguishable, the relationship is referred to as cryptic symbiosis.
2. Commensal invertebrates that live on other invertebrates are called epizoites.
3. Commensal invertebrates that live inside other animals are called endozoites.
4. Organisms that respond positively to stimulation by contact with solid surfaces are said to exhibit thigmotaxis.
5. The unusual feature of the remora is the presence of a suckerlike dorsal fin.
6. Some protozoans ingest algae, but do not digest all of the cellular components; the chloroplasts remain intact, thereby providing the protist with another source of food. This phenomenon is called chloroplast symbiosis.
7. Symbiotic relationships between luminescent bacteria and fishes are best described as mutualistic because both symbionts benefit.
8. Fishes being cleaned must remain motionless, with their mouths open during the cleaning.

178 Test Bank for Marine Biology

Essay Discussion Questions

Develop complete answers to these questions.

1. Discuss the modern view of the endosymbiotic theory of the origin of eukaryotic cells. How does this theory relate to endosymbiosis in coral-reef invertebrates?
2. Discuss the similarities between chloroplast symbiosis and the endosymbiotic theory of the origin of eukaryotic cells.
3. Speculate on the origin of algal symbiosis in invertebrates. How could it have begun, and when did it become stabilized?
4. What anatomical modifications occur in symbiotic dinoflagellates? Speculate on the adaptive significance of these changes.
5. Discuss two behavioral changes that have occurred in the invertebrate part of an algal symbiosis.
6. The epizoic entoproct, *Loxosomella fauveli*, lives in the respiratory channels of the polychaete worm, *Aphrodite aculeata*. How could one tell whether this association is a matter of habitat-dependent survival or habitat selection by the larval stage of the entoproct?
7. Support or refute the following metaphor: The giant clam, *Tridacna gigas*, is a gardener. Discuss.
8. Discuss the mechanisms whereby algal symbionts are transmitted from one generation of host invertebrate to the next.
9. Empty gastropod shells are often in short supply for hermit crabs. Describe how the bryozoan *Hipporidra* eliminates the need for the hermit crab to exchange its shell for a larger one.
10. How are anemone fish capable of hiding in the stinging tentacles of their hosts without being stung? Provide, in your essay, a brief review of the experimental evidence that supports this view.
11. What is teleology, and why is teleological thinking an easy trap into which scientists may fall? Give at least three examples of teleological thinking.
12. In what ways are the symbiotic algae passed from one generation to the next?
13. What happens to the coral–algae symbiosis when the coral are starved? Discuss.
14. How do anemone fish avoid being captured by their host?
15. What happens to the coral–algae symbiosis when the water temperature is elevated? Present evidence to support your answer.
16. What epizoic animals are commonly seen at the seashore? Epiphytic animals?
17. Why are cleaning shrimp (or fish) so brightly colored or otherwise conspicuous?
18. How does the anemone fish prevent itself from becoming the prey of its anemone host? Contrast this behavior with that of *Nomeus* and the Portuguese man-of-war.
19. What is coral bleaching? Discuss this phenomenon in relation to global warming.
20. What is meant by the term “thigmotaxis,” and what does this concept have to do with symbiotic relationships? What adaptive advantage does this behavior provide a symbiont?
21. Speculate as to why there are no algal–vertebrate symbiotic relationships.
22. Develop an experimental protocol to test the idea that the shrimp fish, *Aeoliscus strigatus*, is attached to long, thin rods positioned vertically in the water and not to some other agent or characteristic of the long-spined sea urchin, *Diadema*.
23. Discuss how thigmotaxis may preadapt invertebrates to establish symbiotic relationships with other animals.
24. How can the radiolarian *Acanthometra* function as if it were a plant, producing more oxygen than it consumes via respiration?
25. Discuss the gains that symbiotic algae receive as symbiotes within their hosts.
26. Discuss the range of morphological modifications that animals have evolved within their symbiotic relationships.

27. Discuss the range of physiological modifications that animals have evolved within their symbiotic relationships.
28. How might the fact that the radiolarian *Acanthometra* possesses endosymbiotic zooxanthellae in its cytoplasm influence its distribution in the sea? Discuss.
29. What happens when symbiotic algal cells are cultured outside of their animal host? Speculate on the physiological meaning of this phenomenon.
30. Challenge Question: Speculate as to why there are fewer symbiotic relationships in freshwater than marine systems.
31. Challenge Question: Speculate on the evolution of cleaning behavior in terms of the mathematical model known as the prisoner's dilemma, or the ultimate game.
32. Challenge Question: Invent a symbiotic relationship between a nonphotosynthetic, nonbioluminescent bacterial species and one of the species of invertebrates that you have studied. Explore the gains that each symbiont would experience in this hypothetical relationship, and explain how the relationship is reestablished in each new generation of the host.

